

PAPER

Context-Aware Stock Recommendations with Stocks' Characteristics and Investors' Traits

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SUMMARY Personalized stock recommendations aim to suggest stocks tailored to individual investor needs, significantly aiding the financial decision making of an investor. This study shows the advantages of incorporating context into personalized stock recommendation systems. We embed item contextual information such as technical indicators, fundamental factors, and business activities of individual stocks. Simultaneously, we consider user contextual information such as investors' personality traits, behavioral characteristics, and attributes to create a comprehensive investor profile. Our model incorporating contextual information, validated on novel stock recommendation tasks, demonstrated a notable improvement over baseline models when incorporating these contextual features. Consistent outperformance across various hyperparameters further underscores the robustness and utility of our model in integrating stocks' features and investors' traits into personalized stock recommendations.

key words: recommendation, stock recommendation, investor modeling, behavioral economics, natural language processing, graph neural network

1. Introduction

The number of retail investors using online brokers has been rapidly increasing due to the lower fees and the convenience of trading compared with traditional full-service human brokers. While online brokers provide retail customers with an easy-to-use interface, affordable fees for trading, automated non-personalized recommendations, they failed to provide carefully personalized investment or trading advice that used to be conveyed by professional brokers. These personalized investments advice are valuable, especially in investments as the objective of building a portfolio varies by each retail investors. For example, some investors might solely seek high returns, while others may prefer a portfolio they are more familiar with over a portfolio with more profitability. Therefore, building a personalized stock recommendation system for individual investors is important.

The research on stock recommendation can be classified into a non-personalized approach and a personalized approach [1]–[14]. Most research falls into the non-personalized approach [2]–[4]. Stock recommendations in a non-personalized approach focus on identifying optimal strategies for selecting portfolios that are likely to be more profitable in the future. Few studies tackle the problem of personalized stock recommendations due to the difficulties in collecting data from privacy concerns [1], [5], [6].

Among a few studies, one method of achieving a personalized stock recommendation is to personalize the recommendation based on an individual investor's risk tolerance and a stock's risk-return. Robin *et al.* [1] estimates the investor's risk tolerance from users' portfolios and recommends stock based on the relevance of the stock's risk-return with the user's risk tolerance combined with a collaborative filtering method.

The past research in personalized stock recommendation models faces two technical challenges. The first challenge lies in considering the characteristics of stocks. While previous research primarily focused on assessing the risk-return profile of a stock based on historical data, individual investors take into account multiple aspects when making investment decisions [15], [16]. In other words, individual investors don't rely solely on historical financial time series; instead, they analyze a wealth of information from diverse sources such as text data. To incorporate the various data investors refer to, we specifically focus on the following three factors in financial decision makings.

Technical Indicator

Individual investors evaluate investments and identify trading opportunities by analyzing the historical price pattern in technical analysis. Technical analysis relies on the analysis of historical price and volume data to identify patterns, trends, and signals that could help predict future price movements.

Fundamental Factors

Individual investors assess the financial strengths and management of a company by examining related economic and financial indicators in fundamental analysis. Fundamental analysis focuses on evaluating the intrinsic value of an asset by examining various factors related to the underlying financial, economic, and qualitative aspects of the asset.

Business Activities

Individual investors consider the business activities of a company by examining text data such as earning summaries and news data as it is related to how the stock is influenced by economic events and news. It has been shown that investors' behaviors are influenced by economic events described in textual data such as news and earning reports [15]–[18].

Therefore, it is important to take the three factors into consideration in personalized stock recommendation models, although the previous studies in personalized stock rec-

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ommendation have not considered those traits due to the difficulties in incorporating them. While technical indicators and fundamental factors are relatively easy to utilize for stock recommendation tasks as these features are numerical in nature, it is quite difficult to incorporate the business activities of a stock as they are shown in unstructured data such as text data and network data and not numerical. Therefore, the values of incorporating the stock context information from unstructured data is worth exploring.

The second shortcoming is an insufficiency of personalization on individual investors' traits. Personalizing the stock recommendations for individual investors is important as the objective of building a portfolio varies by each retail investor. The past research personalizes the recommendation only based on one variable: investors' risk tolerance. However, it is unlikely that one variable could capture the complex nature of investors. For example, in the literature on investor modeling, it has been empirically shown that psychological factors such as personality traits and behavioral biases have a non-negligible effect on investor behavior as well as risk tolerance [19]–[22]. Therefore, it is not sufficient to personalize a stock recommendation only based on investors' risk tolerance. Also, the risk tolerance is computed from the investors' portfolio, which makes it impossible to recommend a stock to investors with no past transaction, bringing about the cold-start problem where no prior events are known to certain investors. Therefore, we need to explore the value of investors' take psychological traits in order to model the complex nature of investors.

In response to the above necessity, we analyze the benefits of context-aware recommendations in personalized stock recommendations. First, we quantify the characteristics of the stock. As for technical indicators and fundamental factors, we calculated the most representative indicators and factors. As for business activities, we adopted a stock embedding method to extract features from unstructured data. Second, we quantify the traits of investors. We conducted a questionnaire survey and gathered investors' psychological traits. Third, we verify the advantages of incorporating the contextual information into Neural Collaborative Filtering (NCF) [23].

The main contributions of this paper are summarized as follows.

1. We propose a framework to incorporate context information that enables integrating structured data and unstructured data for personalized stock recommendations. The results of the experiments using NCF show our proposed method outperformed baseline models on the new stock recommendation task.
2. We conduct an ablation study to analyze how incorporating the context of stocks contributes to higher performance.
3. We conduct experiments with different parameters and analyze how parameters will influence the performance of the proposed model.

2. Related Work

2.1 Related works for Personalized Recommendation

This section reviews the previous works in the personalized recommendation. First, we review the basic techniques in a personalized recommendation, and then we review how neural networks are utilized in a personalized recommendation.

2.1.1 Basic Techniques in Personalized Recommendation

Personalized recommendation approaches can be divided into two broad categories: content-based methods and collaborative filtering methods. The content-based methods generally identify the common characteristics of items from users' past ratings and then recommend to a user unseen items that share similar characteristics [24], [25].

On the other hand, collaborative filtering approaches use ratings from other users and items. The key principle is that the ratings of a target user for unseen items are likely to be similar to other users who have similar ratings to the target users. Collaborative filtering approaches can be divided into two general classes of neighborhood methods and model-based methods. In neighborhood-based collaborative filtering methods, the user-item rating matrix is directly used to estimate ratings for unseen items [26]–[28]. For example, Konstan *et al.* [29] assess the interest of a target user for an item using the ratings for the same item from other users, which they called neighbors, that have similar ratings for items. On the contrary, model-based collaborative filtering methods learn a predictive model from the user-item interaction matrix.

2.1.2 Neural Network in Recommendation

Recently, the neural network has been widely used in the studies of personalized recommendations. Neural networks can be used to learn the predictive model for ratings. Standard collaborative filtering methods such as matrix factorization use dot products to model user-item interactions. Deep learning can be used as an alternative to those methods. For example, He *et al.* [23] models the user-item interaction with multilayer perceptron. This work takes the concatenation of a user vector and an item vector or the element-wise multiplication of a user vector and an item vector and uses them as input for multilayer perceptrons.

Neural networks can also be used to extract content representations. For example, text descriptions of items are one of the most used text data for recommender systems. Many studies use deep learning methods to extract feature representations from text and utilize them for recommendations [30]–[32].

2.2 Related Works for Stock Recommendation

There is a growing demand for stock recommendations as

the number of retail investors using online brokers has been rapidly increasing. Accordingly, many studies have tackled stock recommendation tasks. The approach of stock recommendations can be classified into two approaches: non-personalized stock recommendations and personalized stock recommendations. Most works in stock recommendation fall within the scope of non-personalized stock recommendation which focuses on identifying optimal strategies for selecting stocks or portfolios that are likely to be more profitable in the future [4]. On the other hand, little research is done on personalized stock recommendations due to the lack of open data and difficulties in collecting the data from privacy concerns. While there is not much literature on personalized stock recommendations, there are some studies that tackle the problem of personalized stock recommendations. Collaborative filtering has been used for personalized stock recommendation, oftentimes combined with other recommendation approaches such as order book analysis, and multiple criteria decision analysis [1], [5], [7]. For instance, Robin *et al.* [1] estimates the investor's risk tolerance from users' portfolios and recommends stock based on the relevance of the stock's risk return with the user's risk tolerance combined with a collaborative filtering method. The method of personalizing stock recommendations based on investors' risk tolerance has two shortcomings. First, the consideration of stocks' characteristics is insufficient as it only considers the risk-return of stocks. Second, the personalization of investors' traits is not enough. While the past studies con-

sider risk-tolerance of investors, it is not clear whether one variable, risk tolerance can capture the complex nature of investors. Therefore, this paper aims to overcome the two shortcomings.

3. Method

In this section, we provide an overview of our proposed model. As shown in Fig. 1, the model consists of the following three steps. Step 1 extracts the characteristics of stocks such as technical indicators, fundamental factors, and business activities of stock. Technical indicators and fundamental factors are calculated from financial time series and financial statements. We extract the information on business activities using the stock embedding methods [33]. Step 2 extracts the traits of investors such as personality traits, behavioral traits, and investors' attributes. Step 3 integrates the above information and builds neural collaborative filtering methods and predicts the investors' interaction with stocks.

3.1 Stock Characteristics

First, we consider the stock characteristics. Let i be a stock. Then, technical indicators and fundamental factors of stock i at time t are calculated as follows.

$$\text{technical}_{t,i} = TC(v_{t,i} \dots v_{t-n,i}) \quad (1)$$

$$\text{fundamental}_{t,i} = FC(w_{t,i}) \quad (2)$$

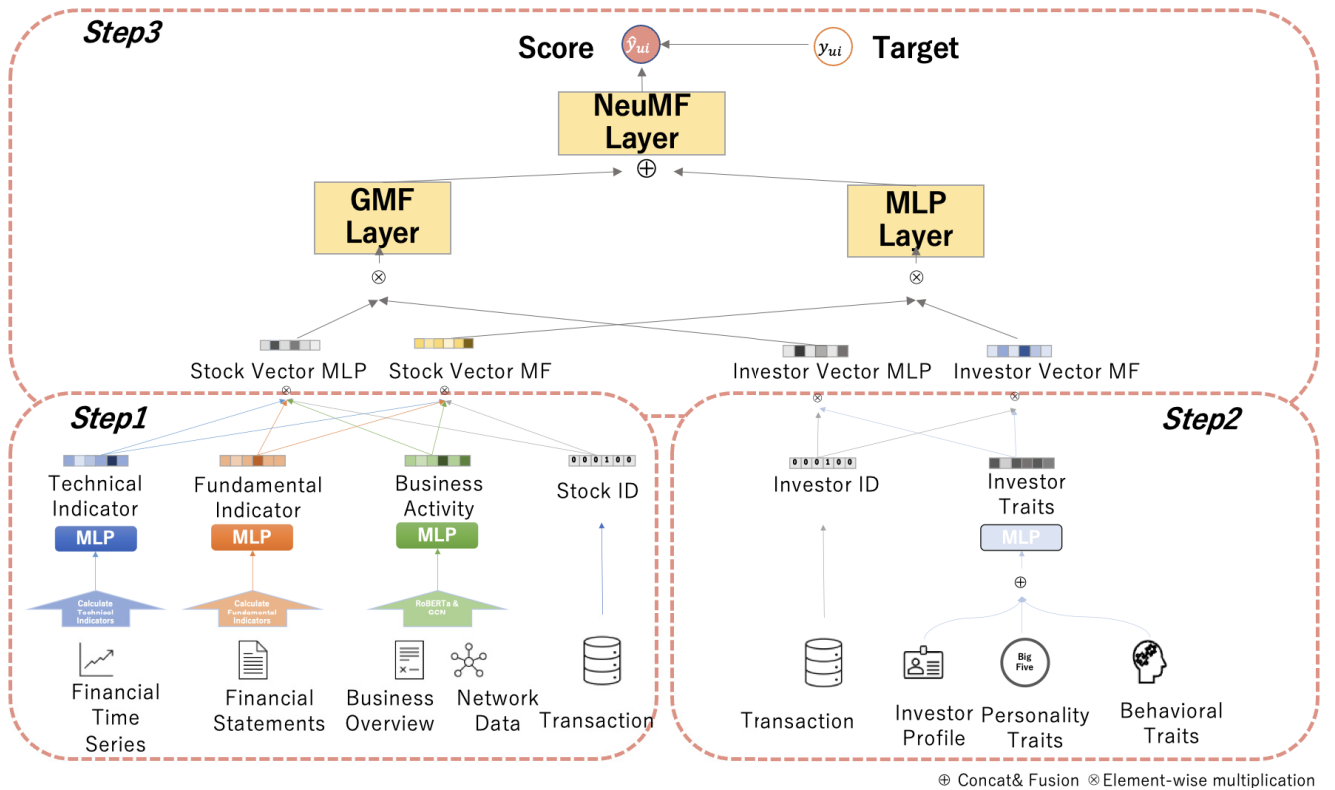


Fig. 1 Outline of our proposed personalized stock recommendation model

where $v_{t,i}$ is a close price of stock i at time t , n is the necessary time period to calculate the technical indicators, $w_{t,i}$ is a vector of financial information such as net sales and profit in financial statements of stock i at time t , TC is a function to calculate technical indicators such as Bollinger bands, FC is a function to calculate fundamental factors such as earnings price share, and $\text{technical}_{t,i} \in \mathbb{R}^{k_t}$, and $\text{fundamental}_{t,i} \in \mathbb{R}^{k_f}$, are the vector of technical indicators and fundamental factors for stock i at time t . k_t and k_f are the dimensions of $\text{technical}_{t,i} \in \mathbb{R}^{k_t}$ and $\text{fundamental}_{t,i} \in \mathbb{R}^{k_f}$ respectively, and k_t is the number of technical indicators to use times the periods to consider, and k_f is the number of fundamental factors to use times the periods to consider.

To capture the business activities of stocks, we employ stock embedding enhanced with textual information and network information (SETN) [33]. SETN combines Bidirectional Encoder Representation from Transformers (BERT) [34] and Graph Convolutional Networks (GCN) [35]. SETN has been proven effective in capturing not only the industry and sector information of stocks but also theme information, making it useful for understanding the business activities associated with each stock. Following the work in [33], stock embeddings are calculated as follows.

$$h_{\text{BERT},i} = \text{BERT}(x_i) \quad (3)$$

$$h_{\text{GCN},i} = \text{ReLU} \left(\sum_{j \in N_i} \hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} h_{\text{BERT},j} W + b \right) \quad (4)$$

$$\text{emb}_i = h_{\text{BERT},i} + h_{\text{GCN},i} \quad (5)$$

where x_i is the text description of target stock i , specifically the business descriptions of the company in annual reports, $h_{\text{BERT},i} \in \mathbb{R}$ is the pooled output of BERT, N_i is neighboring stocks of i , $\hat{A} = A + I$ denotes the adjacency matrix of the graph constructed by causal chain [36] with inserted self-loops and $\hat{D}_{i,i} = \sum_{j=0} \hat{A}_{ij}$ its diagonal degree matrix, W is a weight matrix, b is a bias, and $S_{\text{emb},i}$ is a stock embedding of stock i .

Then, we use an MLP layer to set the stock characteristics to be the same dimension.

$$S_{\text{technical},t,i} = \text{ReLU}(W_{\text{tech}} \text{technical}_{t,i} + b_{\text{tech}}) \quad (6)$$

$$S_{\text{fundamental},t,i} = \text{ReLU}(W_{\text{fund}} \text{fundamental}_{t,i} + b_{\text{fund}}) \quad (7)$$

$$S_{\text{emb},i} = \text{ReLU}(W_{\text{emb}} \text{emb}_i + b_{\text{emb}}) \quad (8)$$

where $W_{\text{tech}} \in \mathbb{R}^{l \times k_t}$, $W_{\text{fund}} \in \mathbb{R}^{l \times k_f}$, $W_{\text{emb}} \in \mathbb{R}^{l \times k_e}$, b_{tech} , b_{fund} , $b_{\text{emb}} \in \mathbb{R}^l$, and l is the dimension size for the latent embedding. Therefore the stock vector q_i is calculated as follows.

$$q_i^{\text{MF}} = S_{\text{technical},t,i} \odot S_{\text{fundamental},t,i} \odot S_{\text{emb},i} \odot p_i \quad (9)$$

$$q_i^{\text{MLP}} = [S_{\text{technical},t,i} \odot S_{\text{fundamental},t,i} \odot S_{\text{emb},i} \parallel p_i] \quad (10)$$

where p_i is an item embedding obtained by applying an embedding layer to one-hot vector of stock i , $\odot$ shows Hadamard product, $\parallel$ shows concatenation operation. In this

formula, we first calculate the Hadamard product followed by a concatenation operation. q_i^{MLP} is the input of the latent stock vector for the multilayer perceptrons (MLP) layer, and q_i^{MF} is the input of the latent stock vector for matrix factorization (MF) layer.

3.2 Investor Traits

Then, we consider the investor traits. Let u be an investor, and $I_{\text{Attributes},u}$, $I_{\text{Personality},u}$, $I_{\text{Behavioral},u}$ be investor u 's attributes, personality traits, and behavioral traits respectively. Investor traits I_u are shown as follows.

$$I_u = [I_{\text{Attributes},u} \parallel I_{\text{Personality},u} \parallel I_{\text{Behavioral},u}] \quad (11)$$

where $\parallel$ shows the concatenation operator. The investor vector r_u is calculated as follows.

$$r_u^{\text{MF}} = I_u \odot p_u \quad (12)$$

$$r_u^{\text{MLP}} = [I_u \parallel p_u] \quad (13)$$

where p_u is a user embedding obtained by applying an embedding layer to a one-hot vector of investor u . r_u^{MLP} is an input of the latent investor vector for MLP layer, and r_u^{MF} is an input of the latent investor vector for MF layer.

3.3 Collaborative Filtering

Lastly, we recommend stocks using collaborative filtering methods. Following the work of [23], we compute the predicted preference $\hat{y}_{u,i}$ as follows.

$$\phi^{\text{GF}} = q_i^{\text{MF}} \odot r_u^{\text{MF}} \quad (14)$$

$$\phi^{\text{MLP}} = a_L(a_{L-1}(\dots a_2(W_2[q_i^{\text{MLP}} \parallel r_u^{\text{MLP}}] + b_2) \dots) + b_L) \quad (15)$$

$$\hat{y}_{u,i} = \sigma(W_F[\phi^{\text{GF}} \parallel \phi^{\text{MLP}}] + b_F) \quad (16)$$

where σ is a sigmoid function, a_l is an activation function such as ReLU, W_l and W_F is a weight matrix and b_l and b_F is a bias term.

3.4 Loss

We define the loss function L as follows.

$$\begin{aligned} L = & - \sum_{(u,i) \in \mathcal{Y}} \log \hat{y}_{u,i} - \sum_{(u,i) \in \mathcal{Y}^-} \log(1 - \hat{y}_{u,i}) \\ = & - \sum_{(u,i) \in \mathcal{Y} \cup \mathcal{Y}^-} y_{u,i} \log \hat{y}_{u,i} + (1 - y_{u,i}) \log(1 - \hat{y}_{u,i}) \end{aligned} \quad (17)$$

where $\mathcal{Y}$ are positive samples, $\mathcal{Y}^-$ are negative samples, and $y_{u,i} \in \{0, 1\}$ is an implicit feedback of investor u to stock i . As for negative samples $\mathcal{Y}^-$, we uniformly sample negative stocks from a previously non-purchased set of stock for each investor.

4. Dataset

In this section, we describe the dataset that we used in our

experiments. We deal with three types of data: stocks' characteristics, investors' traits, and transaction data.

4.1 Stocks' Characteristics

As individual investors refer to technical indicators and fundamental factors for investing decision makings, we calculated the technical indicators and fundamental factors most often used by individual investors for each stock. As for technical indicators, we calculated the five most used technical indicators by individual investors. We calculated the Bollinger band, moving average convergence divergence, relative strength index, average directional index, and stochastics using TA-Lib[†] library which is a Python library to perform technical analysis in the financial market. The Bollinger band is a volatility indicator that shows how volatile the market is by calculating the moving average and standard deviation given a certain period. Moving average convergence divergence is a momentum indicator that shows the relationships between the two moving averages such as 26 days exponential moving averages and 12 days exponential moving averages. The relative strength index is also a popular momentum indicator that works as a metric for price changes and the speed at which they punctuate for a certain period. The average directional index is used to assess both the momentum and the direction of price movement by calculating the minus and plus directional indicators. Stochastic is used to compare the current close price of a stock with a certain period. As for fundamental factors, we calculated the four most used fundamental factors by individual investors. We calculated earnings per share, book value per share, price-to-earnings ratio, and return on equity. Earnings per share are the portion of a company's profit that is assigned to each share of its stock. Book value per share is a ratio of common equity divided by the number of outstanding shares, which is a company's stock currently held by all its shareholders. Price-to-earnings ratio measures the relationship between the stock price of a company and its per-share earnings. It is calculated by dividing the current per share of stock by the stock's earnings per share. Return on equity is a measure of a company's financial performance calculated by dividing net income by average shareholders' equity.

In addition to the technical and fundamental factors, investors also consider the business activities of a company, shown in unstructured data such as earning summaries and news data. In order to capture that information, we use the stock embedding method: SETN [33] which is shown to be effective in capturing stocks' characteristics from unstructured data such as textual data and network data. In learning stock embedding, we utilized the CoARiJ (Corpus of Annual Reports in Japan)^{††}, which consists of organized Japanese financial reports. This corpus encompasses annual reports, integrated reports, corporate social respon-

sibility (CSR) reports, and earning summaries from 3,016 companies, spanning the years 2014 to 2018. During the experiments, we utilized a company's business description, which provides an overview of the company's activities, to embed textual information.

4.2 Investors' Traits

We collected investors' psychological traits such as personalities, behavioral biases, cognitive ability, and goals of investment. For personality data, we collected the Big-Five personality traits, openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism, using a ten-item personality inventory (TIPI) [37], [38]. For behavioral data, we collected risk preference, time discount, overconfidence, hyperbolic discounting, sign effect, and magnitude effect. When designing the questionnaire and calculating investors' traits, we referred to the questionnaire used in the Japan Household Panel Survey (JHPS)^{†††}. Risk preference shows how much investors prefer taking risks. Time discount is a current relative valuation placed on receiving money at an earlier date compared with receiving it at a later date. Overconfidence is a behavioral bias of the tendency for an investor to overestimate their abilities of trading. Hyperbolic discounting is a tendency to value immediate but smaller rewards more than long-term larger rewards. Sign effect is the steeper discounting of gains compared to losses. Magnitude effect is the tendency that preferences are sensitive to the sizes of the rewards being considered. For cognitive ability, we measured financial literacy by asking questions about financial knowledge and wealth management, while we used a cognitive reflection test to obtain investors' cognitive ability [39]. Lastly, we asked about the goals of investments. The options include retirement, housing, education, medical, vacation, and others. As for investors' attributes, we utilized investors' attributes information such as age, annual income, amount of financial assets, experiences in investment, and investment attitudes.

4.3 Transaction Data

We collected 969 investors' transaction data from a Japanese securities company. The data ranges from July 2020 to September 2022. We processed the transaction data into a user-item matrix given m users and n items. Following the work in [1], we define $m \times n$ matrix U_f with components

$$(U_f)_{ij} = f(i, j) \quad (18)$$

Let $q_{i,j,t}$ be the portfolio of user i on stock j on the day t which is obtained from transaction data.

$$q_{i,j,t} = \begin{cases} 1 & \text{if user } i \text{ holds stock } j \text{ in time } t \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

We define implicit feedback collaborative filtering user-item matrix R as U_{f_R} in Eq. (20).

[†]<https://github.com/TA-Lib/ta-lib-python>

^{††}<https://github.com/chakki-works/CoARiJ>

^{†††}https://www.iser.osaka-u.ac.jp/survey_data/survey_eng.html

$$f_R(i, j) = \begin{cases} 1 & \text{if there is } t \in T \text{ s.t. } q_{i,j,t} \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

where T is an entire period. Simply speaking, rows of the R matrix represent, for user i , whether they held stock j at a certain period.

5. Experiments

In this section, we conduct 3 experiments with the aim of answering the following research questions.

Research question 1 (RQ1) Does our proposed personalized stock recommendation model outperform the baseline models in stock recommendation tasks?

Research question 2 (RQ2) How does added data help improve the model's performance?

Research question 3 (RQ3) Is the result robust to the change in hyperparameters?

We first explain the experimental settings, then we answer the above research questions.

5.1 Experimental Settings

5.1.1 Tasks

We evaluated our experiments in the new stock recommendation task to assess the ability to identify the relevant stocks for a given user. In new stock recommendations, we remove the repeat consumptions of stocks and extract the transactions when the investors buy new stocks that have not been seen in their portfolios. Then, we process the data into the user-item matrix and build a recommendation model to recommend the top K items to each user. This task can be interpreted as evaluating the performance of recommending new stocks to investors. In behavioral finance, it has been empirically shown that investors tend to stick to familiar stocks rather than exploring the full universe of assets, and the tendency is called "familiarity bias" [40]. Therefore, it is meaningful for stock recommender systems to recommend new stocks to investors which might help mitigate familiarity bias by exploring the universe of options. We used HR@K, and NDCG@K for evaluation metrics.

5.1.2 Evaluation Protocol

We adopted the widely used leave-one-out evaluation method in order to evaluate the performance of stock recommendation [23]. We held out the latest interaction as the test data for each individual investor. We randomly chose one interaction for each investor and used it as validation data, and we utilized all the remaining data for training data. As it takes too much time to rank all the items for every evaluation, we followed the commonly-used strategy [23] that randomly samples 100 stocks that have not interacted with users in the past, and ranked the stock in test data with 99 items. We truncated the ranked list at 10 for both metrics

following the work in [23]. Intuitively, HR assesses whether the test item is in the top-10 recommended list, and the NDCG accounts for the position of the hit by assigning the higher scores when it hits at a higher rank. We calculated both metrics for each user and reported the average score.

5.1.3 Baselines

We compared our proposed personalized stock recommendation model with the following methods. These baseline models capture the interactions of users and items from past transactions, while they do not utilize side information.

1. **Generalized Matrix Factorization (GMF)**. This method is the generalized method of matrix factorization (MF), with is the most popular model for a recommendation [41].
2. **Multi-Layer Perceptron (MLP)**. This is simple model to use multi-layer perceptron. This model first concatenates the user one-hot vector with the stock one-hot vector and use several layers of MLPs.
3. **Neural Collaborative Filtering (NCF)**. This model uses both GMF and MLP and is one of the most used collaborative filtering methods using Neural Networks [23].

5.1.4 Parameter Settings

We implemented our proposed models using Pytorch. As for the first and second experiments, we set the latent dimension l to 100. Following the work in [23], we used four layers of MLP. The architecture of MLP layers is $2l \rightarrow 4l \rightarrow 2l \rightarrow l$, noting that the first layer is $2l$ as it is the concatenation of two vectors with dimension l . We used the Adam optimizer with a learning rate equal to 0.001 for the first and second experiments.

5.1.5 Experiments

We have conducted three experiments to address the above three research questions.

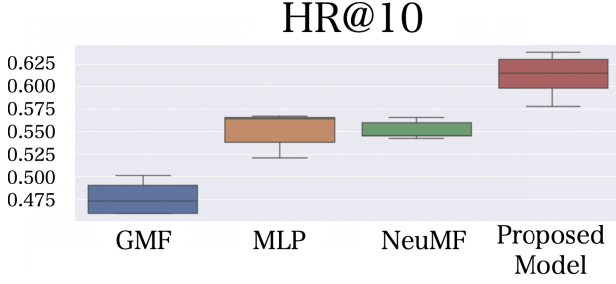
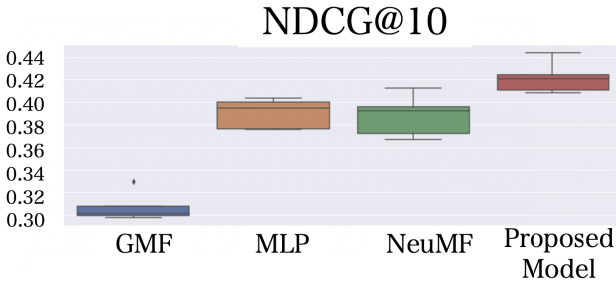
Experiment 1 To address RQ1, we conducted the first experiment to compare the performance of our proposed model with that of the baseline model.

Experiment 2 To address RQ2, we conducted the second experiment to investigate the values of added data to the performance by conducting the ablation study of the combinations of stock characteristics such as technical indicators, fundamental factors, and business activity.

Experiment 3 To address RQ3, we conducted the third experiment to assess whether the proposed model is robust to the changes in hyperparameters. We compared the performance of the model on a parameter when the latent dimension $l = [64, 128, 256, 512]$ and the learning rate = [0.01, 0.001, 0.0001, 0.00001].

Table 1 Results for The First Experiment

	HR@10	NDCG@10
Model		
GMF	0.476	0.307
MLP	0.551	0.390
NCF	0.551	0.388
Proposed Model	0.612	0.422

**Fig. 2** Boxplot of results of HR@10 with 5 seed values for the first experiment. The whisker shows the minimum and the maximum. The box shows the first quartile, median, and third quartile.**Fig. 3** Boxplot of results of NDCG@10 with 5 seed values for the first experiment. The whisker shows the minimum and the maximum. The box shows the first quartile, median, and third quartile.

Experiment 1: to address RQ1, we conducted the first experiment to compare the performance of our proposed model with that of the baseline model.

Experiment 2: to address RQ2, we conducted the second experiment to investigate the values of added data to the performance by conducting the ablation study of the combinations of stock characteristics such as technical indicators, fundamental factors, and business activity.

Experiment 3: to address RQ3, we conducted the third experiment to assess whether the proposed model is robust to the changes in hyperparameters. We compared the performance of the model on a parameter when the latent dimension $l = [64, 128, 256, 512]$ and the learning rate $= [0.01, 0.001, 0.0001, 0.00001]$.

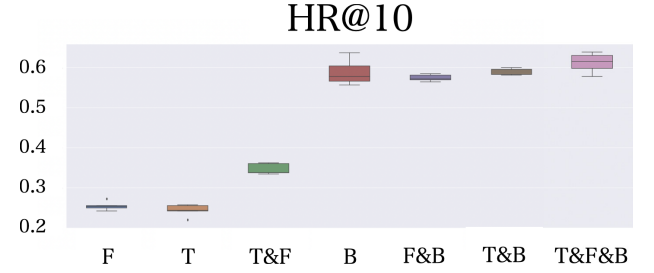
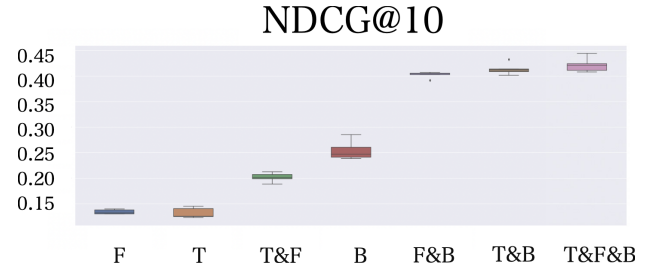
5.2 Results

5.2.1 Result of Experiment 1 (RQ1)

The result for the first experiment is shown in Table 1. We evaluated our models with five different seeds and reported the average of the scores. Figure 2 and Fig. 3 are the box

Table 2 Results for The Second Experiment

	NDCG@10	HR@10
Stock Characteristics		
Technical	0.133	0.249
Fundamental	0.132	0.256
Business	0.246	0.576
Technical & Fundamental	0.198	0.342
Technical & Business	0.415	0.588
Fundamental & Business	0.401	0.576
Technical & Fundamental & Business	0.421	0.605

**Fig. 4** Boxplot of results of HR@10 with 5 seed values for the second experiment. The whisker shows the minimum and the maximum. The box shows the first quartile, median, and third quartile.**Fig. 5** Boxplot of results of NDCG@10 with 5 seed values for the second experiment. The whisker shows the minimum and the maximum. The box shows the first quartile, median, and third quartile.

plots of HR@10 and NDCG@10 for each model with different seeds.

5.2.2 Result of Experiment 2 (RQ2)

The result for the second experiment is shown in Table 2. In this experiment, we evaluated our models with five different seeds and reported the average of the scores. Figure 4 and Fig. 5 are the box plots of HR@10 and NDCG@10 for each model with different seeds, where T, F, and B stand for technical, fundamental, and business activity respectively.

5.2.3 Result of Experiment 3 (RQ3)

The result for the third experiment is shown in Table 3 and Table 4. Table 3 reports HR@10 score of the experiment, and Table 4 reports NDCG@10 score of the experiment.

6. Discussion

For addressing RQ1, we experimented to analyze whether

Table 3 HR@10 of The Third Experiment

Model	#dim of latent layer	Learning rate		
		0.001	0.0001	1.00E-05
GMF	64	0.507	0.117	0.098
GMF	128	0.478	0.099	0.099
GMF	256	0.476	0.099	0.099
GMF	512	0.507	0.094	0.099
MLP	64	0.515	0.479	0.479
MLP	128	0.578	0.484	0.555
MLP	256	0.584	0.521	0.510
MLP	512	0.570	0.575	0.470
NCF	64	0.535	0.488	0.436
NCF	128	0.549	0.488	0.545
NCF	256	0.598	0.553	0.536
NCF	512	0.598	0.573	0.467
Our Model	64	0.601	0.576	0.529
Our Model	128	0.624	0.544	0.562
Our Model	256	0.589	0.593	0.582
Our Model	512	0.636	0.636	0.573

Table 4 NDCG@10 of The Third Experiment

Model	#dim of latent layer	Learning rate		
		0.001	0.0001	1.00E-05
GMF	64	0.322	0.041	0.044
GMF	128	0.307	0.053	0.044
GMF	256	0.298	0.040	0.044
GMF	512	0.322	0.041	0.044
MLP	64	0.360	0.293	0.291
MLP	128	0.419	0.328	0.341
MLP	256	0.414	0.370	0.327
MLP	512	0.413	0.420	0.324
NCF	64	0.386	0.289	0.272
NCF	128	0.417	0.341	0.339
NCF	256	0.452	0.383	0.325
NCF	512	0.435	0.401	0.321
Our Model	64	0.410	0.372	0.320
Our Model	128	0.440	0.355	0.353
Our Model	256	0.414	0.394	0.367
Our Model	512	0.447	0.443	0.362

the proposed model outperforms the baseline models. The result shows that the proposed model outperforms the baseline models of GMF, MLP, and NCF in both metrics. The result in Table 1, Fig. 2, and Fig. 3 shows that our proposed models consistently outperformed the baseline models with different seeds. Therefore, we confirmed our hypothesis to enhance the recommendation with stocks' characteristics and investors' traits work for stock recommendation tasks. To understand how the stock embeddings obtained by SETN can effectively capture related stocks, we conducted an analysis of the three closest neighboring stocks for the selected stock, as shown in Table 5. While TOPIX33 is a commonly used index in the Japanese market to represent industrial sectors, TOPIX33 has been reorganized into TOPIX17 for enhanced convenience. For instance, we chose Kintetsu Group Holdings Co., Ltd as the selected company, and the following three companies were identified as the nearest neighbors based on the embeddings. Notably, the embeddings successfully capture the business activities of both the selected company and the nearest companies operating in similar business areas. These examples illustrate how SETN's embeddings help personalized stock recommendations based on a company's business activities.

For addressing RQ2, we conducted an ablation study on data types. The result in Table 2 shows that the model maximized the performance when all types of technical, fundamental, and stock embeddings are utilized. Also, the result in Fig. 4, and Fig. 5 shows that the models with all the data outperformed other models consistently with different seeds. Looking at the graduate performance changes for results in Table 2, we observe three findings. First, the stock embedding itself has brought high performance in HR@10, while the result of NDCG@10 is relatively low. Second, the technical indicators and fundamental factors themselves do not bring either high HR@10 or high

Table 5 Three nearest neighbors for stock embedding using SETN

Company	TOPIX17	TOPIX33
Kintetsu Group Holdings Co.,Ltd	Transportation & Logistics	Land Transportation
TOBU RAILWAY CO.,LTD.	Transportation & Logistics	Land Transportation
Kyushu Railway Company	Transportation & Logistics	Land Transportation
KobeElectricRailway Co., Ltd	Transportation & Logistics	Land Transportation
NIPPON EXPRESS CO., LTD	Transportation & Logistics	Land Transportation
Inui Global Logistics Co., Ltd	Transportation & Logistics	Marine Transportation
Kintetsu World Express, Inc	Transportation & Logistics	Warehousing and Harbor Transportation
Nippon Yusen Kabushiki Kaisha	Transportation & Logistics	Marine Transportation
FUJITSU GENERAL LIMITED	Electric appliances & Precision instruments	Electric Appliances
DAIHEN Corporation	Electric appliances & Precision instruments	Electric Appliances
HOKUETSU INDUSTRIES CO., LTD	Machinery	Machinery
NISSIN ELECTRIC Co.,Ltd	Electric appliances & Precision instruments	Electric Appliances
SBI Holdings, Inc	Financial (ex banks)	Securities and Commodities Futures
Mizuho Leasing Company, Limited	Financial (ex banks)	Other Financing Business
MONEY PARTNERS GROUP CO.,LTD	Financial (ex banks)	Securities and Commodities Futures
TOYO SECURITIES CO.,LTD	Financial (ex banks)	Securities and Commodities Futures

NDCG@10 both in individual cases and a combined case. We consider this because technical indicators and fundamental factors alone are not the main factors for individual investors' decision makings in general. Thus, combining either technical indicators or fundamental factors with interactions worsens performance. Especially, in Eq. (12) where r_u^{MF} takes Hadamard product of context information and item embedding, the technical indicators can be thought to work as noise for the model. Third, when the technical indicators or fundamental factors are added to the stock embedding it brings higher performance in both metrics, especially in NDCG@10. These observations can be understood that stock embedding broadly predicts the test stock in the top ten lists, while it fails to assign the test stock a higher rank. Then, technical indicators and fundamental factors help to assign test stock a higher rank to the recommended list. This corresponds to the investor behavior that investors first choose the candidate stocks to buy by looking at companies' business activity. Then they look at the financial time series of the stock or examine the financial situation of the company to decide what to buy in detail.

For addressing RQ3, we compared the performance of the model with different parameters. We compare the proposed models with baseline models with different latent dimension size and learning rates. The result shows that the proposed model outperforms the baseline models consistently with different hyperparameters in both metrics.

7. Conclusion

This study addresses the issues in personalized stock recommendation. Especially, we tackle the challenges of incorporating various stock characteristics and investors' traits. To address these challenges, we first quantify the stocks' characteristics that investors refer to in investment decision makings. We especially consider technical indicators, fundamental factors, and business activities. Then, we quantify the investors' traits using the data collected from a questionnaire survey. We especially consider personality traits, behavioral traits, cognitive ability, the goal of investment, and attribute information. Finally, we show that a context-aware stock recommendation model enhanced with stocks' characteristics and investors' traits outperformed the baseline models in the new stock recommendation task using HR@K and NDCG@K. The result shows that considering stocks and investors' context contributes to the performance of personalized stock recommendations.

We plan to extend this work to build an explainable personalized stock recommendation model. In financial regulations, it is important to state the reasons why the model recommends a stock to an investor. While our proposed model succeeds in personalizing recommendations based on stocks and investors' context, our model could not provide interpretability in recommendations. Therefore, we plan to incorporate an attention mechanism to visualize the important weights of how much investors value each stock's characteristics. Also, we plan to conduct a user study to verify

whether the interpretability provided by the attention mechanism aligns with the actual investors' interest.

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